Lianming Power – Digital Power Modules

CAN Communication Protocol (External)

Revision History

Date of revision	Version	Description	Reviser
2021-04-20	V1.0	First draft	Dai Chang
2022-09-19	V2.0	Add Precautions	Tang Jianjun

1. Overview

This agreement employs CAN specification version 2.0B, utilising a 29-bit identifier and extended frames.

2. Scope of Application

This agreement describes the CAN communication protocol for external use by Shenzhen Lianming Power Supply Co., Ltd. It applies to power modules manufactured by Lianming Power Supply, including the 10kW series, 12kW series, 20kW series, 30kW series charging modules, and laser power supplies.

3. Physical Communication Interface

Baud rate: 125 kbit/s.

4. Frame structure

The 29-bit identifier is defined as follows:

28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Byte3					Byte2								Byte1								Byte0							
Command ID_H																						Module Address						

Command Parsing 1:

Name	Command ID	Module Address	Data length	Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
Monitoring Output Settings	(0x1907, C080)>>7	0~60	8	CMD=0	Current (mA)*1000			Voltage (mV)*1000			
Module return output	(0x1807, C080)>>7	1~60	8	CMD=0	NZ, NZ: Current and voltage settings succeeded						
					0, NZ: Current failed, voltage succeeded						
					NZ, 0: Current successful, voltage failed						
					0, 0: Current and voltage settings failed						
Read module information	(0x1907, C080)>>7	1~60	8	CMD=1	/	/	/	/	/	/	/
Module return status	(v0x1807, C080)>>7	1~60	8	CMD=1		Current (A*10)		Voltage (V*10)		Status 1	Status 0
Set Module Power On/Off	(0x1907, C080)>>7	0~60	8	CMD=2	/	/	/	/	/	/	0x55 Power on 0xAA Power off
Module Power On/Off Response	(0x1807, C080)>>7	1~60	8	CMD=2	NZ, NZ: Setup succeeded 0, 0: Setup failed						

Address 0 indicates a broadcast command, which is executed by all modules on the bus. Broadcast commands do not generate a response from the modules.

Addresses 1–60 indicate individual module addresses.The module with the corresponding address returns the response data.

NZ indicates non-zero data (default 0xFF). In the data field, the high byte is transmitted first, followed by the low byte.

If a module does not receive a monitoring command within 2 minutes, communication is considered interrupted.

"/" indicates reserved and shall be filled with 0.

Module status flag bit definitions are as follows:

Byte7:Bit	7	6	5	4	3	2	1	0
Define Status 0	1: Output under-voltage 0: Output normal	1: Output overvoltage 0: Output normal	1: Input under-voltage 0: Input normal	1: Input overvoltage 0: Input normal	1: Fan fault 0: Fan normal	1: Module constant current 0: Module constant voltage	1: Module fault 0: Module normal	1: Module Off 0: Module Running
Panel indicator light	1: Yellow Light On	1: Red Light On	1: Red Light On	1: Red Light On	1: Red Light Blinking	1: Yellow Light On	1: Red Light On	

Byte6:Bit	7	6	5	4	3	2	1	0
Define the state 1						1: Set shutdown 0: Set startup	1: Over-temperature protection 0: Module normal	1: Overcurrent protection 0: Module normal
Panel indicator light						1： Green Light Blinking 0： Green Light On	1: Red light On	1: Red light On

Note: For the above alarms, except when the panel indicator Light is Red On,in which case the module will automatically shut down for protection,the module continues to operate normally for other alarms.

Command Parsing 2:

Name	Command ID	Module Address	Data length	Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7
Read the monitoring set voltage	(0x1901, 0080)>>7	1~60	0	Not used							
Module returns voltage value	(0x1801, 0080)>>7	1~60	8							Set voltage（V*10）	
Read Monitoring Current Limit Setting	(0x1901, 0880)>>7	1~60	0	Not used							
Module Return Current Limit	(0x1801, 0880)>>7	1~60	8	Current limiting value（A*10）							
Read Input Voltage Value	(0x1907, A080)>>7	1~60	8	0x31	/	/	/	/	/	/	/
Module Return Input Voltage	(0x1807, A080)>>7	1~60	8	0x31		Vab/32 (Input single-phase module Vac/32）		Vbc/32		Vca/32	
Read Environment Temperature Value	(v0x1900, 8080)>>7	1~60	0	Not used							
Module Return Environment Temperature	(0x1800, 8080)>>7	1~60	8					Environment Temperature（° C*10）			
Module Current Sharing Control	(0x19C2, 1880)>>7	0	6	/	/	/	0x55 Close module Autonomous current sharing 0xAA Enable module current sharing	/	/		
Module Internal Current Sharing	(0x1807, 8080)>>7	1~60	8	The module automatically reports current sharing data every 400 ms, and no user attention is required.							
Module Address Lookup	(0x19C2, 2880)>>7	1~60	8	/	/	/	0x55 Module Lookup Green light blinking for 10 seconds	/	/	/	/

5. Message Example

1) . Set broadcast on/off

Command ID	1907C080	Data :	2	0	0	0	0	0	0	55
Return ID		Data :								

Broadcast commands without return

2) . Set Module 1 Power On/Off

Command ID	1907C081	Data :	2	0	0	0	0	0	0	55
Return ID	1807C081	Data :	2	FF	0	0	0	0	0	0

Set successfully: Byte1 returns FF; failure returns 00

3) . Set the output voltage and current for Module 3

Command ID	1907C083	Data :	0	01	5F	90	00	01	86	A0
Return ID	1807C083	Data :	0	FF	0	0	0	0	0	0

Module 3 Output Configuration 100V,100A

100V = 100*1000 = 100000mV = 0x186A0

90A = 90*1000 =90000mA = 0x15F90

Set successfully: Byte1 returns FF; failure returns 00

4). Read the status information of Module 2

Command ID	1907C082	Data :	1	0	0	0	0	0	0	0
Return ID	1807C082	Data :	1	0	01	17	03	E8	0	0

Output voltage value: Byte4~Byte5
Output current value: Byte2~Byte3
Output status: Byte6~Byte700

0x03E8 = 1000/10 = 100V

0x0117 = 279/10 = 27.9A 00

Operating normally, no alarms

5) . Read Monitoring Output Voltage of Module 1

Command ID	19010081	Data :								
Return ID	18010081	Data :	1	0	18	36	0	1E	03	84

Output voltage value set for Module 1: Byte6~Byte7 0x0384 = 900/10 = 90V

6) . Read Monitoring Output Current Limit of Module 5

Command ID	19010885	Data :								
Return ID	18010885	Data :	01	3D	0	0	0	0	40	2C

Output current limit value set for Module 5: Byte0~Byte1

0x013D = 317/10 =31.7A

7) . Read Input Voltage of Module 1

Command ID	1907A081	Data :	31	0	0	0	0	0	0	0
Return ID	1807A081	Data :	31	01	2F	1E	2F	2C	2F	26

Input voltage value for Module 1 Vab：

Byte2~Byte3 0x2F1E = 12062/32 =376.9V

Vbc：Byte4~Byte5 0x2F2C = 12076/32 =377.4V

Vca：Byte6~Byte7 0x2F26 = 12070/32 =377.2V

8) . Read Environment Temperature of Module 1

Command ID	19008081	Data :								
Return ID	19008081	Data :	0	1F	0B	B5	01	52	03	A1

Environment temperature: Byte4~Byte5 0x0152 = 338/10 =33.8°C

9) . Disable Module Autonomous Current Sharing

Command ID	19C21880	Data :	0	0	0	55	0	0		
Return ID		Data :								

Byte4 = 0x55 disables current sharing, 0xAA enables current sharing;

10) . Module Address Lookup

Command ID	19C22882	Data :	0	0	0	55	0	0	0	0
Return ID	19C22882	Data :								

Locate Module 2. The module at address 2 will flash its green light for 10 seconds.

6. Precautions

1) The module must not be loaded directly during soft start. The load can only be applied after the module output voltage has been established (steady-state output). This can be determined via the module status flag bits (see Table 6-1 and Figure 6-1).

Byte7:Bit	7	6	5	4	3	2	1	0
Define State 0	1: Output under-voltage 0: Output normal	1: Output overvoltage 0: Output normal	1: Input under-voltage 0: Input normal	1: Input overvoltage 0: Input normal	1: Fan fault 0: Fan normal	1: Module constant current 0: Module constant voltage	1: Module fault 0: Module normal	**1: Module Off 0: Module Running**
Panel indicator light	1: Yellow light On	1: Red light On	1: Red light On	1: Red light On	1: Red light Blinking	1: Yellow light On	1: Red light On	

Table 6-1

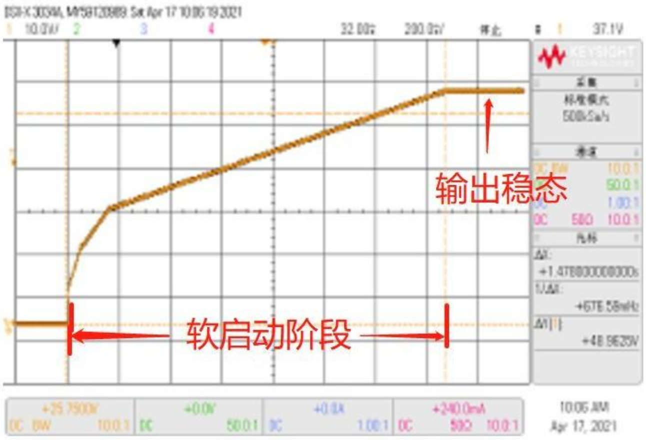

Figure 6-1

2) The module output current setpoint should be increased in a stepwise manner. For example,if the required current is 100 A, the initial setpoint can be 10 A, and then increased in 10% load current, i.e., 10 A, 20 A,…, 90 A, 100 A. The recommended current ramp-up time is ≥1 second.